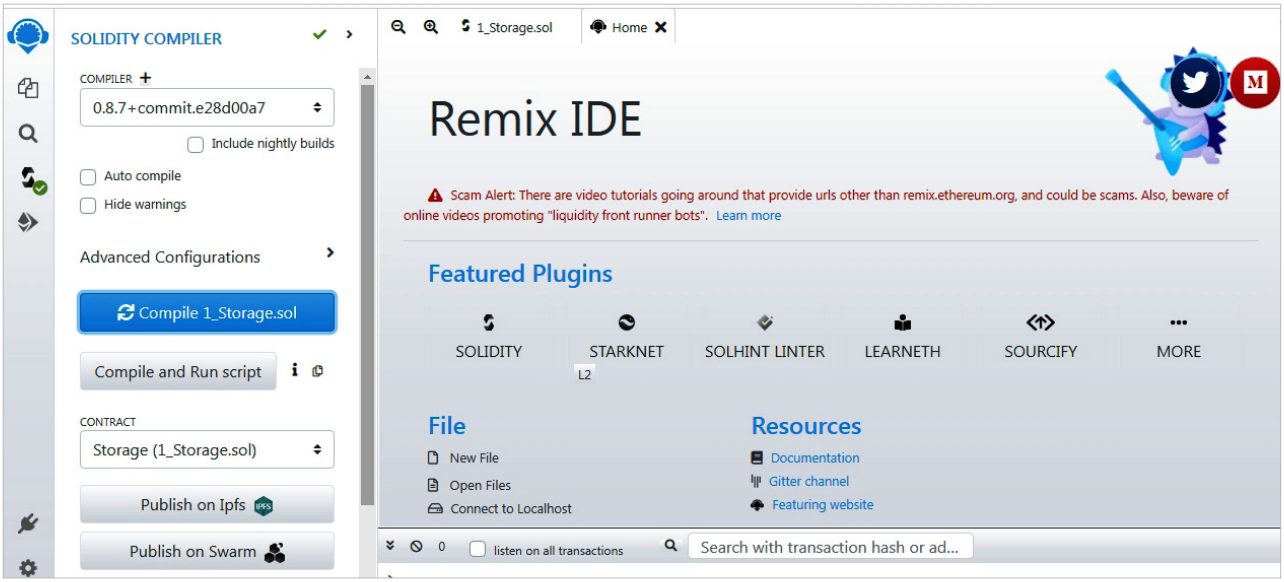

Figure 2: Compilation of Solidity smart contracts in Remix IDE

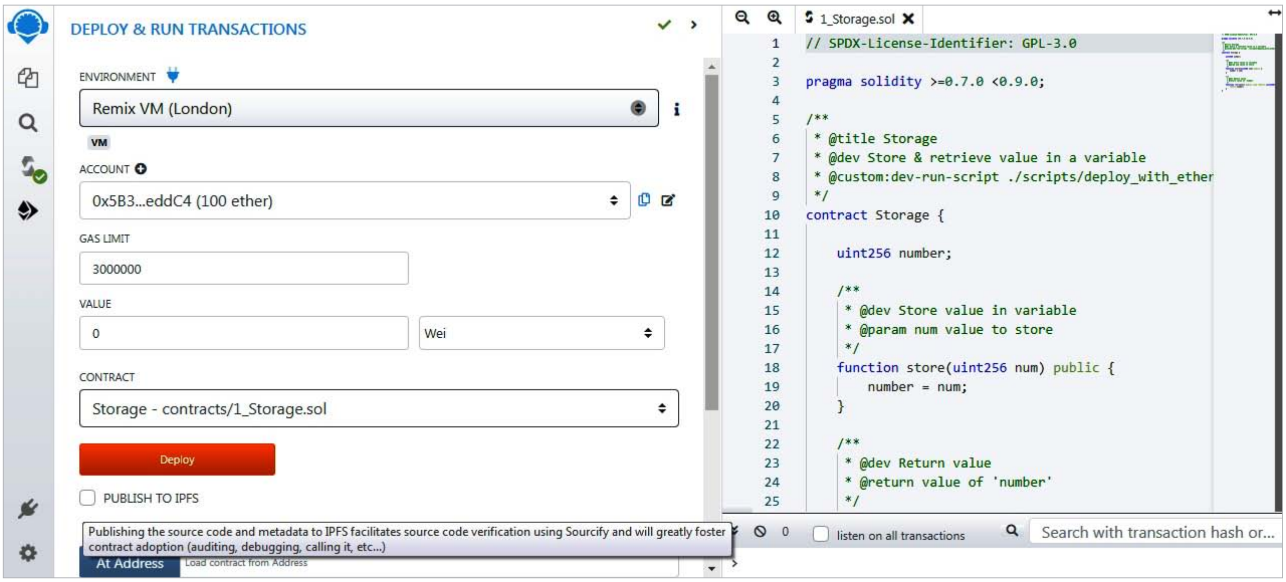

Figure 3: Execution and deployment of smart contracts in Remix IDE

application, which uses smart contracts, in real-time is extremely challenging since the hacker would have to hack all the devices involved with the program. To ensure the safety of transactions, dApps based on smart contracts use a system of dynamic token sharing.

Global transactions necessitate the use of smart contract software. To put it another way, this software allows business to be conducted between persons who otherwise couldn't do so due to geographical distance, language barriers, or cultural differences. Smart contracts allow parties who don't speak the same language to do business without the need for a third party to verify the legitimacy of the transaction.

Solidity programming language for smart contracts

When it comes to developing high-performing smart contracts, one of the most potent options is Solidity, a powerful and fast programming language. It is compatible with many different blockchain systems and is built on the object-oriented programming paradigm, which provides a greater level of security and performance. Solidity code is converted to bytecode and runs on the Ethereum virtual machine (EVM).

Python, JavaScript, and C++ constitute the backbone of Solidity, allowing it to be adapted to a wide